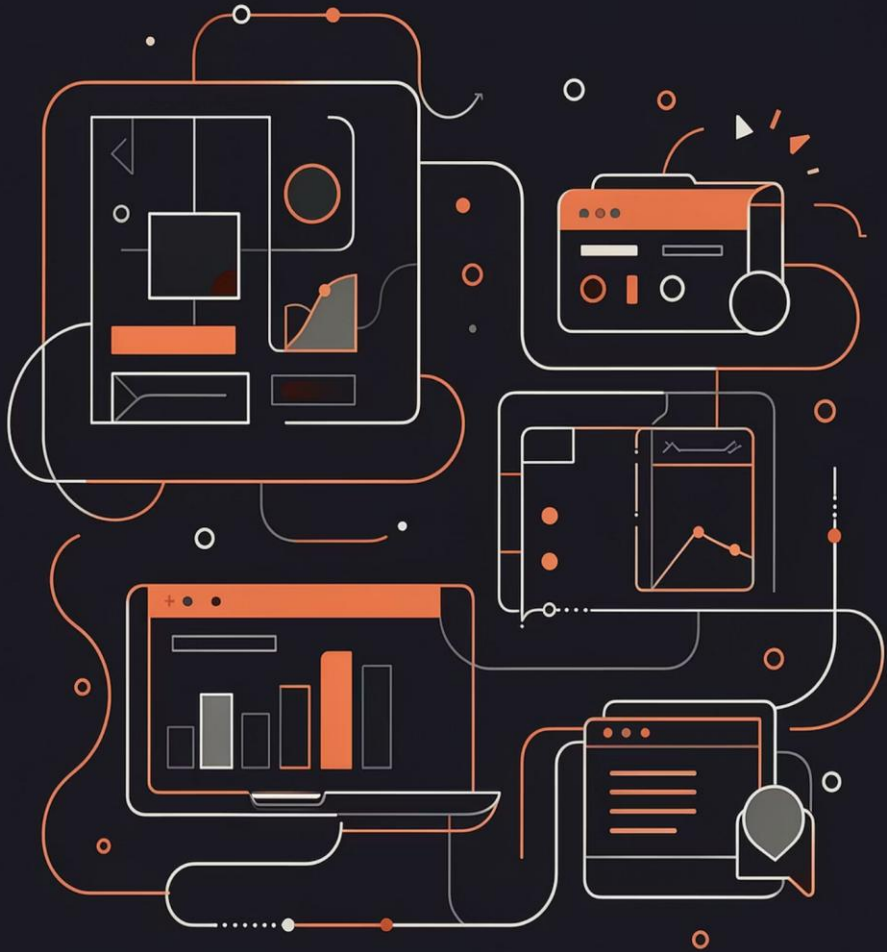


Healthcare Data Cleaning & Exploratory Data Analysis

Name:- *vikas prajapati*

Tools: Python (Pandas, NumPy, Matplotlib, Seaborn)

Date:- 22/04/2026



Introduction

- Data in raw form is often messy and inconsistent
- Data cleaning is a crucial step before analysis
- Exploratory Data Analysis (EDA) helps understand patterns
- This project focuses on cleaning and analyzing healthcare data using Python

Project Objective

- Clean and preprocess raw healthcare dataset
- Handle missing and inconsistent data
- Perform EDA
- Extract meaningful insights from the dataset

Dataset Overview

- The dataset contains healthcare-related records
- Includes patient and hospital information
- Key features:
 - Age
 - Gender
 - Medical Condition
 - Admission Date
 - Billing Amount
 - Insurance Provider
 - medication

Data Cleaning Steps

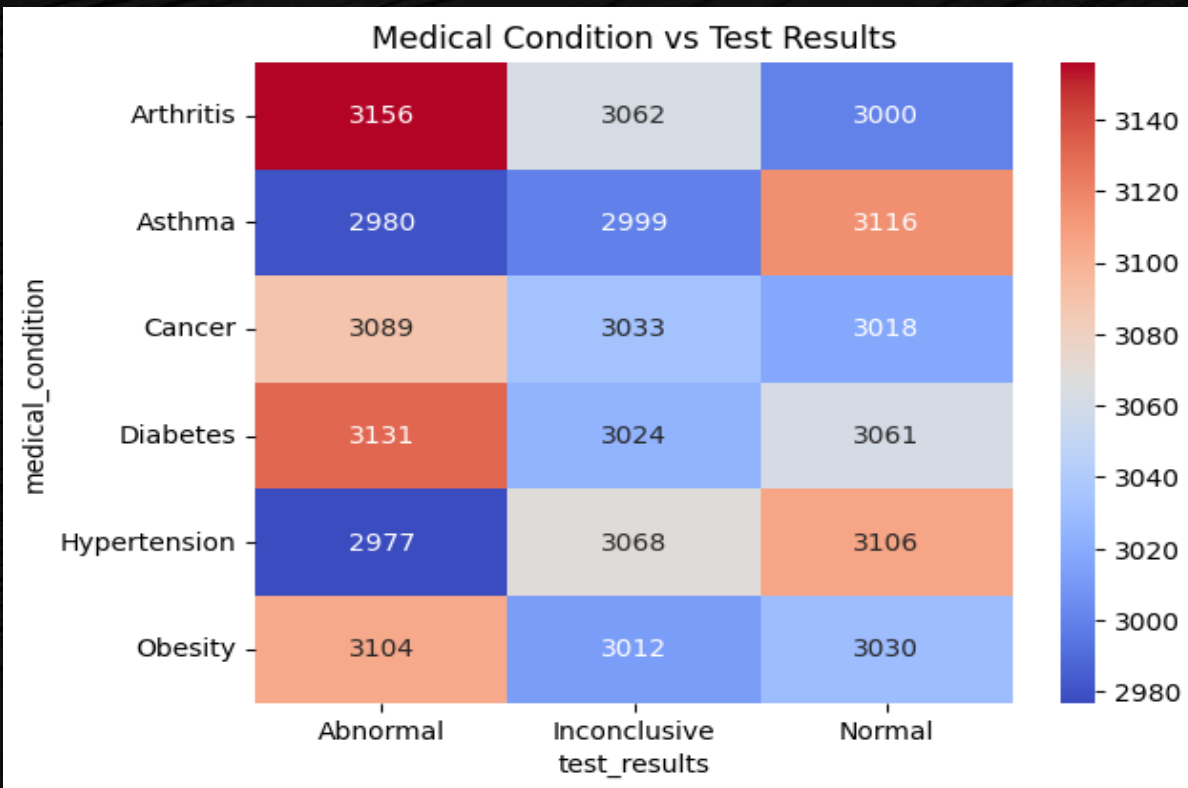
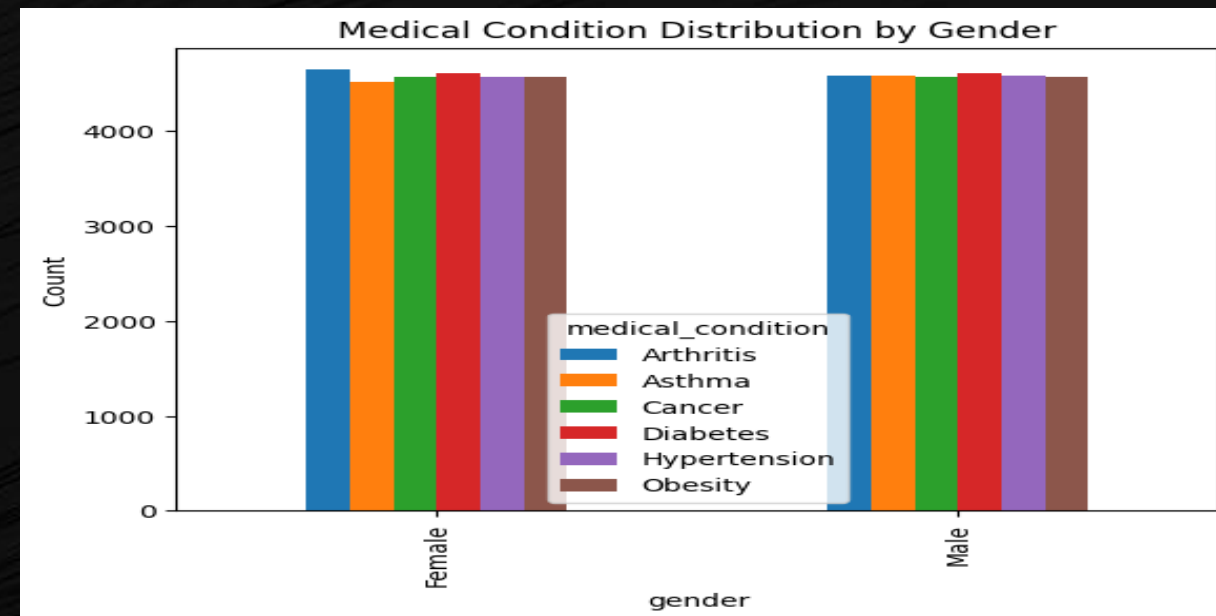
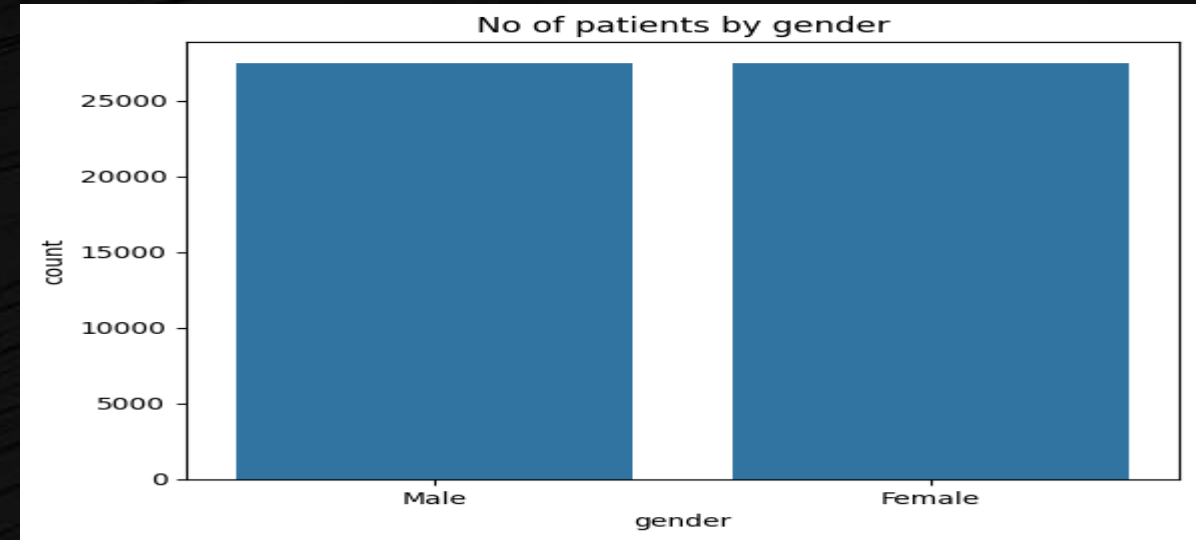
- Removed duplicate records
- Correct name columns (name)
- Converted incorrect data types
- Standardized column names

Exploratory Data Analysis (EDA)

- Analyzed distribution of data
- Identified trends and patterns
- Compared different variables
- Used visualization for better understanding

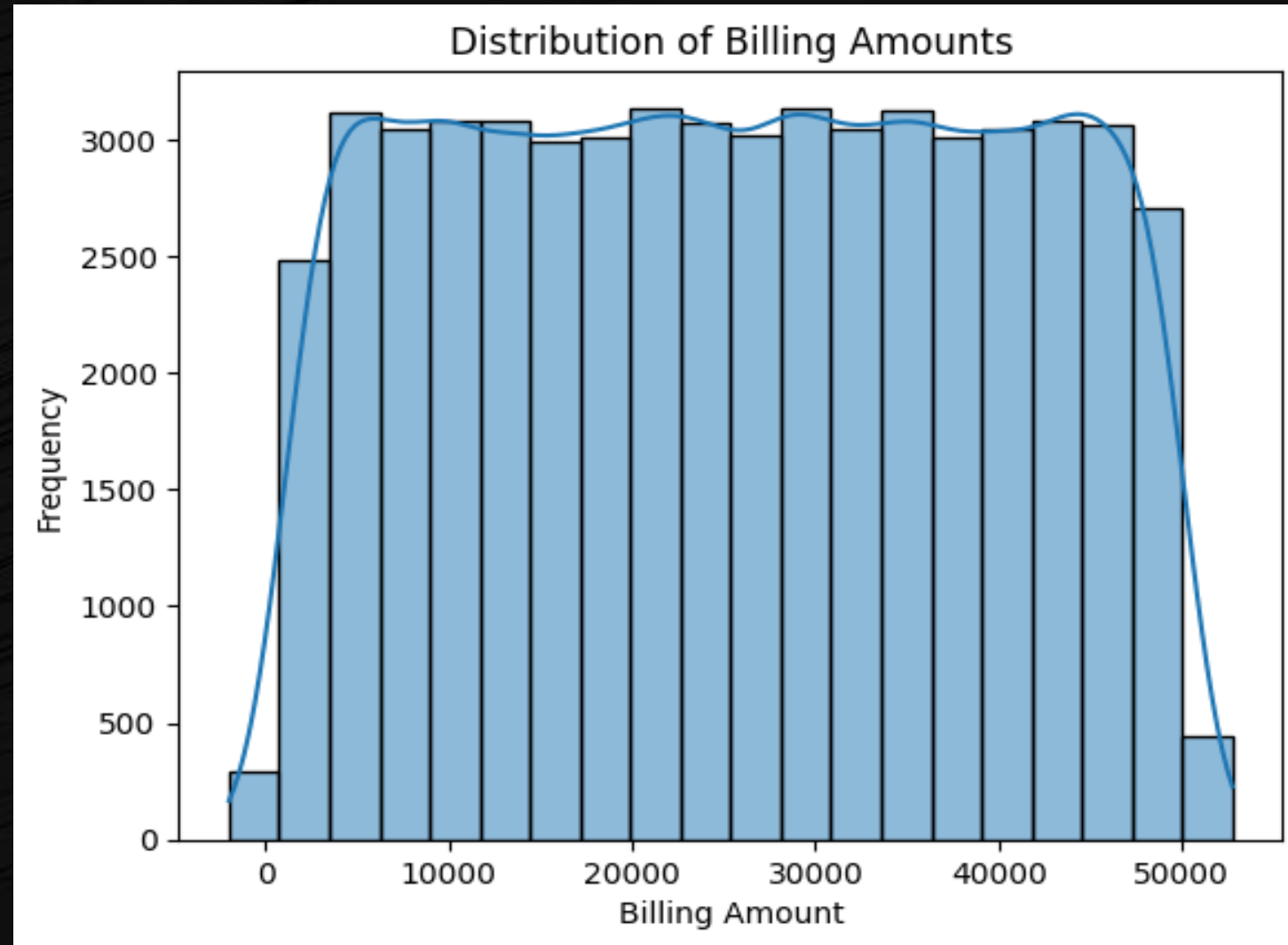
Visualizations Used

- Bar Charts → No patient by gender & Medical condition distribution by gender
- Heat map → Medical condition by test result



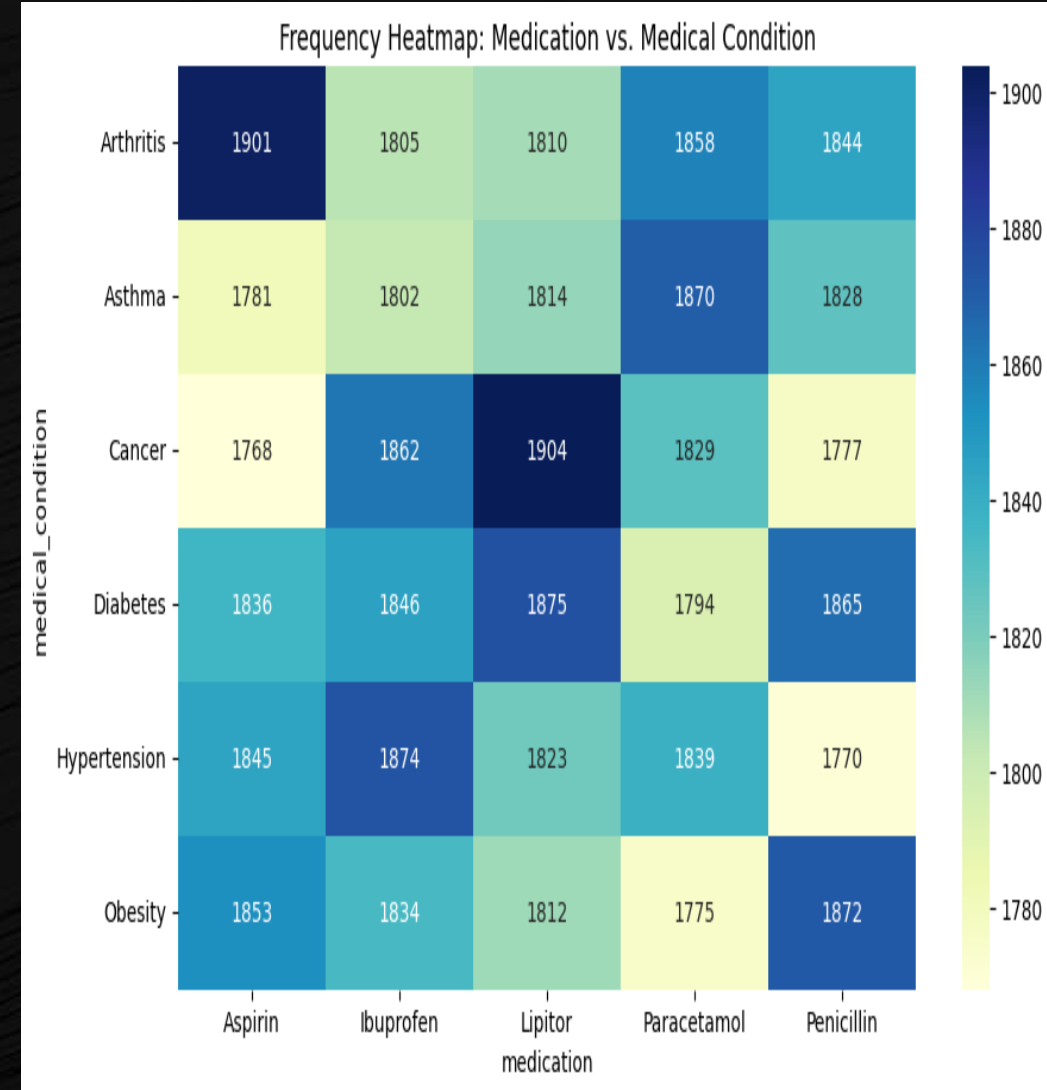
Analysis Performed

- Age distribution of patients
- Gender-wise analysis
- Most common diseases
- Billing amount trends



Key Insights

- Majority of patients belong to age group :- 50 – 65
- **Most Used Medication per Condition:-**
 - **Arthritis** → Highest with *Aspirin* (1901)
 - **Asthma** → Highest with *Paracetamol* (1870)
 - **Cancer** → Highest with *Lipitor* (1904)
 - **Diabetes** → Highest with *Lipitor* (1875)
 - **Hypertension** → Highest with *Ibuprofen* (1874)
 - **Obesity** → Highest with *Penicillin* (1872)
- Billing amount is higher for ____obesity patients
- Data cleaning improved analysis accuracy



Conclusion

- Data cleaning is essential before analysis
- EDA helps uncover useful insights
- Python is effective for data analysis tasks
- Clean data leads to better decision-making